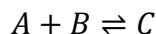


Practice: Reversible Reactions — Oxygen in the Blood

Ungraded practice • No submission required

The Goal: Extend the simple reaction model to a **reversible physiological process** and explore how changing conditions can shift the reaction in different directions.

Consider the reversible reaction:



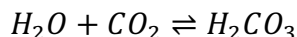
Using the **law of mass action**:

$$\frac{dC}{dt} = k_+[A][B] - k_-[C]$$

where k_+ is the forward rate constant and k_- is the reverse rate constant.

We talked about this in class:

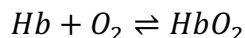
Consider **sparkling water**. Carbon dioxide can dissolve in water and participate in the reversible reaction:



When conditions change, the reaction can move in either direction.

Physiological Example: Oxygen in the Blood

Hemoglobin (Hb) in red blood cells can bind oxygen:



Here:

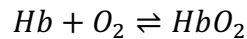
$$A = Hb, B = O_2, C = HbO_2$$

In the **lungs**, oxygen is abundant, so hemoglobin tends to bind oxygen.

In the **tissues**, oxygen is being consumed by cells, so oxygen can dissociate from hemoglobin and become available to the tissues.

1. Think About the Physiology

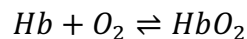
Consider:



- What do A , B , and C represent?
- What do k_+ and k_- represent?
- What happens when $[O_2]$ increases?
- What happens when $[O_2]$ decreases?
- Why is a **reversible reaction** more appropriate here than $A + B \rightarrow C$?
- When might the forward and reverse reaction rates become equal (**equilibrium**)?

2. Implement via AI-Assisted Coding

Use an **AI-assisted coding tool** to implement:



with:

$$\frac{d[HbO_2]}{dt} = k_+[Hb][O_2] - k_-[HbO_2]$$

Simulate how $[Hb]$, $[O_2]$, and $[HbO_2]$ change over time.

Try changing:

- the initial oxygen concentration $[O_2]$;
- the forward rate constant k_+ ;
- the reverse rate constant k_- .

Observe: How do these changes affect oxygen binding and release? Can you relate what you observe in the model to what happens in the **lungs versus the tissues**?